

Jiale Chen

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[Home](#) | [Github](#) | [ORCID](#) | [Google Scholar](#) | [Instagram](#)

Beijing, China

EDUCATION

- **Beijing University of Technology** 2023 - 2027
B.S. in Electrical and Computer Engineering Beijing, China
 - GPA: 3.70/4.00; Overall Average: 88.44/100; Junior Year Weighted Average: 92.7/100 (Top10)
 - Selected Courses:
 - * Advanced Mathematics (100), Control Systems (99), Deep Learning (98), RF Integrated Circuit Design(97), Microcontroller Systems (97), General Physics (97), Digital Integrated Circuit Design (95).
- **Shanghai Jiao Tong University** 2026 - 2027
Research Internship Shanghai, China
 - Designed an end-to-end research framework for medical large-model applications, including data processing, model adaptation, multimodal fusion, privacy protection, and evaluation.
 - Investigated blockchain-based privacy protection for medical large-model inference and fine-tuning strategies for multimodal medical models.
 - Conducted few-shot transfer learning on an orbital imaging dataset and reproduced representative state-of-the-art methods for baseline comparison and framework validation.

PUBLICATIONS

- [1] **Jiale Chen.**
Communication-Efficient Compression Method for FedDyn Federated Learning Models. [🔗]
IEEE 4th International Conference on Control, Electronics and Computer Technology (ICCECT), 2026.
Published in IEEE Xplore.
- [2] Yihong Chen*, **Jiale Chen***, Hongyu Zhao, Lili Cao, Kaili Qin, Yuan Qiu, Ruhui Ma**, Yuan Liu.
MedChain: Blockchain-based Trusted Evidence Preservation for Medical LLM Inference. [🔗]
International Conference on Blockchain and Web3 Technology Innovation and Applications (BWTAC).

INTERNSHIP

- **Lenovo** [🌐][🔗] Jul 2025 - Sep 2025
Embedded Algorithm Engineer Beijing, China
 - Optimized and deployed Stable Diffusion 3 (SD3) for text-to-image, image-to-image, and inpainting tasks, improving inference latency.
 - Built a segmentation-guided image editing pipeline integrating SD3 and SAM2, enabling automatic image segmentation and high-quality inpainting.
 - Leveraged OpenVINO to deploy and accelerate SD3 on Intel platform, achieving efficient inference performance.
- **Cisco** [🌐] Jan 2026 - Feb 2026
Technical Engineer Intern Beijing, China
 - Learns fundamental concepts of AI data center networking, including RDMA over Converged Ethernet (RoCE) and ECN-based congestion control mechanisms.
 - Assists in reviewing switch architectures and network designs for large-scale distributed training scenarios.
 - Supports basic testing and validation tasks for network switches and optical modules under the guidance of engineers.
 - Studies emerging networking technologies and related topics, including programmable networking and reinforcement learning concepts.

RESEARCH EXPERIENCE

- **Communication-Efficient Model Compression for FedDyn in Federated Learning** Dec 2025 - Mar 2026
Advisor: Prof. Ruhui Ma | Shanghai Jiao Tong University 
 - Studied communication efficiency challenges in FedDyn under heterogeneous federated learning settings.
 - Analyzed the trade-off between parameter transmission cost and convergence stability.
 - Developed a FedDyn-aware parameter importance evaluation strategy to enable selective communication, effectively reducing bandwidth usage without sacrificing model performance.
- **FPGA-Based Remote Laboratory Platform** Mar 2025 - Aug 2025
Advisor: Prof. Sujuan Liu | Beijing University of Technology 
 - Designed and implemented a cloud-enabled FPGA remote laboratory system based on FPGA, Raspberry Pi 5, and STM32, enabling full remote experiment workflow.
 - Developed a web-based control platform integrating remote JTAG download, logic analyzer, virtual oscilloscope, digital twin, and real-time IP camera monitoring.
 - Implemented simulated JTAG bitstream download via UART (115200 baud), achieving 10–20s remote deployment latency and stable hardware configuration.
 - Built a real-time signal acquisition and debugging framework using FPGA AD/DA modules and protocol-level analysis (UART/I2C/SPI), enabling remote waveform visualization and system-level validation.
- **Medical Large Language Model** Mar 2026 - ?
Advisor: Hongyu Zhao | Shanghai Jiao Tong University 
 - Studied data preprocess methods and applied them in medical LVLm.
 - Design a blockchain-based system to preserve the privacy during the inference process of Medical Large Language Model.

SKILLS

- **Languages:** IELTS: 7.5 (Listening: 8.0, Reading: 8.5). CET-4: 610. CET-6: 513 (both achieved on the first attempt in freshman year).
- **Programming Languages:** Python, C, C++, Verilog, CUDA.
- **FPGA EDA Tools:** Vivado, QuartusII, PDS.
- **Other EDA Tools:** Virtuoso, Keil5, Modelsim, Proteus, STM32CubeMX.
- **Statistical Software:** SPSS, Lingo, MATLAB.

HONORS AND AWARDS

- **Second Prize, National Undergraduate Integrated Circuit Innovation Entrepreneurship Competition** Aug 2025
Talent Exchange Center of the Ministry of Industry and Information Technology (MIIT), P.R. China 
- **First Prize, Beijing Integrated Circuit Design Competition (Digital Track)** May 2026
Beijing Municipal Education Commission
- **Third Prize, Chinese Collegiate Computing Competition (National Final)** Jul 2026
Organizing Committee of the Chinese Collegiate Computing Competition (4C), P.R. China 
- **Second Prize, China Undergraduate Mathematical Contest in Modeling (Beijing Region)** Nov 2025
China Society for Industrial and Applied Mathematics (CSIAM) 

MISC

- **Hobbies:** Football, Guitar, and Singing.
- I would like to visit different countries and cities, enjoying multiple culture and exploring novel fields.